TMUA / PRACTICE PAPER

Set C Paper 1

Applications of Mathematical Knowledge 20 questions · 75 minutes

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Inequalities

1. For which real values of x is the following inequality true?

$$x^2 < (x-3)^2 < (2x+1)^2$$

A $x < -4$ or $\frac{2}{3} < x < \frac{3}{2}$

B $x < -4$ or $x > \frac{2}{3}$

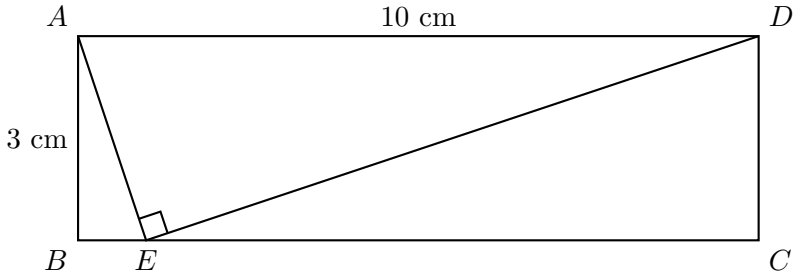
C $-4 < x < \frac{2}{3}$

D $-4 < x < \frac{3}{2}$

E $\frac{2}{3} < x < \frac{3}{2}$ or $x > 4$

Geometry

2. $ABCD$ is a rectangle with $AB = 3$ cm and $AD = 10$ cm.
Point E lies on BC, and $\angle AED = 90°$
What is the value of $BE \times EC$?



A 9 cm^2

B 18 cm^2

C 41 cm^2

D 50 cm^2

E 82 cm^2

Exponentials & Logarithms

3. For $1 < x < 81$, the equation

$$\log_2(\log_3 x) + \log_2\left(\log_3 \frac{81}{x}\right) = 1$$

has two solutions. What is the smaller solution?

A $3^{2-\sqrt{3}}$

B $3^{2-\sqrt{2}}$

C $3^{\sqrt{2}}$

D $3^{4-\sqrt{2}}$

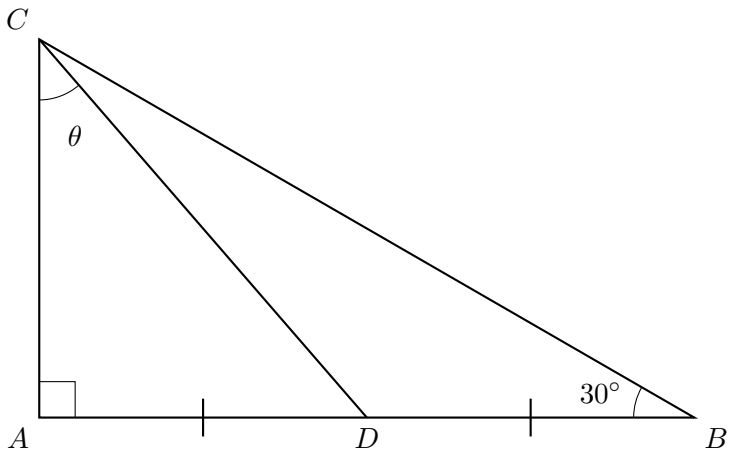
E $3^{2+\sqrt{2}}$

Trigonometry

4. Triangle ABC is right-angled at A, and angle ABC is $30°$

The point D is the midpoint of AB

If angle ACD is θ, what is $\cos\theta$?



A $\dfrac{1}{\sqrt{7}}$

B $\dfrac{1}{2}$

C $\dfrac{\sqrt{3}}{\sqrt{7}}$

D $\dfrac{2}{\sqrt{7}}$

E $\dfrac{\sqrt{3}}{2}$

Sequences & Series

5. An infinite geometric series has positive first term a and common ratio r, where $|r| < 1$. The sum to infinity of this series is S.

A second series is formed by taking every third term of the first series, beginning with the first term:

$$T = u_1 + u_4 + u_7 + u_{10} + \dots$$

where u_n denotes the nth term of the first series.

What is the maximum possible value of $\dfrac{T}{S}$?

A $\dfrac{3}{4}$

B 1

C $\dfrac{7}{6}$

D $\dfrac{4}{3}$

E $\dfrac{3}{2}$

F 3

Exponentials & Logarithms

6. Consider the equation

$$x^{\log_2(x^2)} - 18x^{\log_2 x} + 32 = 0$$

What is the sum of all real solutions to this equation?

A 1

B $\frac{9}{2}$

C 6

D $\frac{25}{4}$

E $\frac{13}{2}$

F $\frac{27}{4}$

G $\frac{31}{4}$

H 18

Differentiation

7. A straight line is tangent to both the curve $y = x^2$ and the curve $y = \frac{1}{x}$, where $x \neq 0$

At what value of x does this line cross the x-axis?

A -2

B $-\frac{3}{2}$

C -1

D $-\frac{1}{2}$

E $-\frac{1}{4}$

Sequences & Series

8. Five positive numbers are consecutive terms of a geometric sequence
The middle three have sum 14 and multiply to make 8
What is the sum of all five numbers?

A 74

B 78

C 80

D 82

E 84

F 86

Probability & Counting

9. A bag contains 3 red balls and 2 blue balls.

Two balls are drawn without replacement.

If they are the same colour, both are returned to the bag. If they are different colours, both are left out.

A ball is then drawn from the bag.

What is the probability that this final ball is red?

A $\frac{3}{5}$

B $\frac{19}{30}$

C $\frac{16}{25}$

D $\frac{2}{3}$

E $\frac{7}{10}$

Algebra

10. Real numbers x and y satisfy

$$x + y = xy$$

and

$$(x-1)^2 + (y-1)^2 = 7$$

What are the possible values of xy?

A -5 and 1

B -3 and 3

C -1 and 3

D -1 and 5

E 1 and 5

Probability & Counting

11. Find the coefficient of x^3y^2 in the expansion of

$$(1 - 2x + y)^8$$

A -17920

B -13440

C -4480

D -560

E 560

F 4480

G 17920

Inequalities

12. The function f is defined for all real x by

$$f(x) = \frac{x^2 + 4x + 1}{x^2 + 1}$$

Which of the following gives all possible values of $f(x)$ as x varies over the real numbers?

A $-1 \leq f(x) \leq 3$

B $f(x) \leq -1$ or $f(x) \geq 3$

C $1 \leq f(x) \leq 3$

D $-3 \leq f(x) \leq 1$

E $-1 \leq f(x) \leq 1$

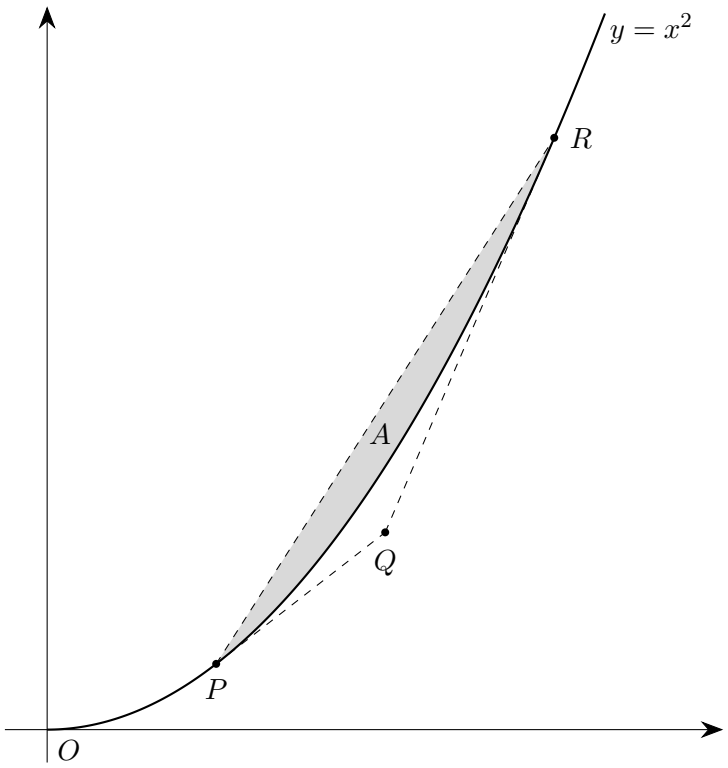
F $f(x)$ can take all real values

Coordinate Geometry

13. The points P and R lie on the curve $y = x^2$ and have x-coordinates 1 and 3, respectively. The tangents to the curve at P and R meet at Q.

The area enclosed by the curve and the line segment PR is A. The area of triangle PQR is B.

What is the value of $\dfrac{A}{B}$?



A $\dfrac{1}{3}$

B $\dfrac{1}{2}$

C $\dfrac{2}{3}$

D $\dfrac{3}{4}$

E $\dfrac{3}{2}$

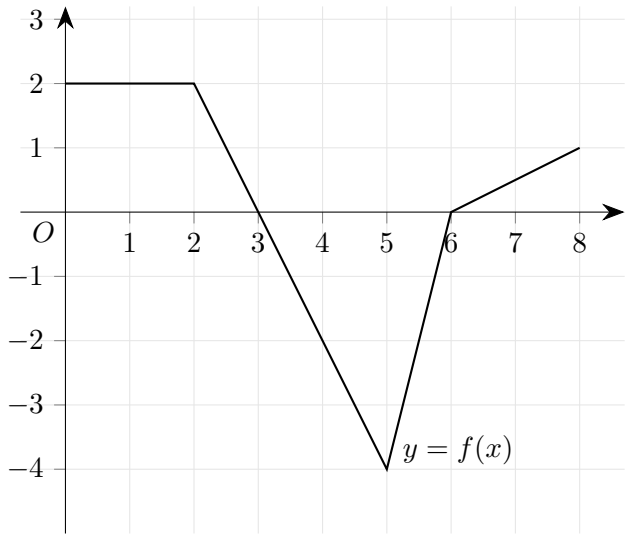
Integration

14. The diagram shows the graph of $y = f(x)$ for $0 \leq x \leq 8$. The graph is made up of straight line segments joining the points $(0, 2)$, $(2, 2)$, $(5, -4)$, $(6, 0)$ and $(8, 1)$.

The function g is defined by

$$g(x) = \int_0^x f(t)\,\mathrm{d}t \qquad \text{for } 0 \leq x \leq 8$$

How many values of x with $0 \leq x \leq 8$ satisfy $g(x) = 0$?



A 0

B 1

C 2

D 3

E 4

F 5

Number

15. Let k be a positive integer, and let $N = 2^4 \times 3^k$

The product of all the positive factors of N is the fourth power of an integer.

What is the smallest possible value of k?

A 1

B 2

C 3

D 5

E 7

F 8

Polynomials

16. A quadratic polynomial f has roots 1 and 3.

The equation $f(f(x)) = 0$ has exactly three distinct real solutions.

What is the value of $f(0)$?

A -9

B -3

C -1

D 3

E 9

Trigonometry

17. A triangle has an angle of $120°$

The perimeter of the triangle is 15, and its area is $\frac{15\sqrt{3}}{4}$

What is the length of the longest side of the triangle?

A 5

B $\frac{11}{2}$

C 6

D $\frac{13}{2}$

E 7

F $\frac{15}{2}$

G 8

Coordinate Geometry

18. The point A has coordinates $(-7, -5)$ and the point B has coordinates $(23, 45)$.

How many points on the straight line segment AB have both coordinates equal to integers, not counting A and B themselves?

A 4

B 5

C 9

D 10

E 11

F 29

G 49

Algebra

19. The positive real numbers x and y satisfy the equations

$$x\sqrt{y} + y\sqrt{x} = 8$$

and

$$x\sqrt{x} + y\sqrt{y} = 40$$

What is the value of $x + y$?

A 6

B 8

C 10

D 12

E 14

F 16

Integration

20. Let f be a polynomial of degree at most 3 with real coefficients.

For $t > 0$, define

$$D(t) = \int_0^t f(x)\,\mathrm{d}x - \int_{-t}^0 f(x)\,\mathrm{d}x$$

Which of the following statements are true?

I If $f(x) = f(-x)$ for every real x, then $D(1) = 0$

II If $D(1) = 0$, then $f(x) = f(-x)$ for every real x

III If $D(1) = D(2) = 0$, then $f(x) = f(-x)$ for every real x

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III